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AI APTITUDE QUESTION BANK

AI Reliability and Evaluation

About This Question Bank

These questions cover the practise of building, testing, and monitoring reliable AI systems: hallucination detection, evaluation frameworks, guardrails, production monitoring, and adversarial safety testing.

5

Sets

15

Questions per Set

75

Total Questions

SET GUIDE

Set 1 · Hallucination Detection

Hallucination detection — taxonomy, entity hallucination, SelfCheckGPT, RAGAs

Set 2 · Evaluation Frameworks

Evaluation frameworks — BLEU, MMLU, LLM-as-judge, golden test sets

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Guardrails and fallbacks — input/output filters, PII, constitutional AI

Set 4 · Monitoring and Testing AI Outputs

Monitoring and testing — production monitoring, A/B testing, chaos testing

Set 5 · Human Oversight and Adversarial Testing

Human oversight and adversarial testing — red-teaming, bias evaluation, safety benchmarks

Each question has one correct answer. Select the best answer from the options provided.

Correct answers and explanations follow each question.

SET

1

Hallucination Detection

Set 1 of 5 · 15 Questions · AI Reliability and Evaluation

Q1. What is the most reliable approach for detecting hallucinations in a RAG system's output?

- A. Measuring the model's token probability scores and flagging outputs with low confidence.
- B. Comparing each claim in the response against the retrieved source documents using an LLM-as-judge evaluation.**
- C. Running the same prompt multiple times and flagging responses that differ across runs.
- D. Checking whether any sentences in the response appear verbatim in the training corpus.

Correct Answer: B**Explanation**

In a RAG system, source documents are available and verifiable. An LLM judge can check: "Is each claim in the response supported by the provided documents?" This provides grounded, systematic hallucination detection. Probability scores are unreliable; variance across runs measures inconsistency, not accuracy; verbatim matching misses correct paraphrases.

Q2. What is a "factual consistency" check and how is it different from a "factual accuracy" check?

- A. Factual consistency checks grammar and style; factual accuracy checks content.
- B. Factual consistency requires human evaluation; factual accuracy can be automated.
- C. Factual consistency applies to structured data; factual accuracy applies to prose responses.
- D. Factual consistency checks whether the response aligns with the provided context; factual accuracy checks whether the response reflects real-world truth.**

Correct Answer: D**Explanation**

Factual consistency (faithfulness) is an internal check: does the response contradict its own retrieved sources? Factual accuracy is an external check: is the response true relative to the real world? A response can be internally consistent (faithful to sources) but factually inaccurate if the sources themselves are wrong. Both matter, but for different reasons.

Q3. What is "SelfCheckGPT" as a hallucination detection technique?

- A. Generating multiple samples from the same prompt and checking whether consistent facts appear across samples — consistent facts are likely true; inconsistent ones are likely hallucinated.**
- B. Asking the model to check its own output for errors before returning the response.
- C. A benchmark dataset for evaluating language models on factual accuracy tasks.
- D. A prompt engineering technique that instructs the model to flag uncertain statements.

Correct Answer: A

Explanation

SelfCheckGPT leverages a key insight: a model is more likely to be consistent about things it "knows" and inconsistent about things it is fabricating. By sampling several outputs from the same prompt and identifying claims that vary across samples, SelfCheckGPT detects probable hallucinations without a reference — useful when no ground truth is available.

Q4. What is "hallucination taxonomy" and why does it matter for mitigation?

- A. A list of topics on which models are known to hallucinate, used to pre-filter queries.
- B. A structured dataset of known hallucination examples used to evaluate detection systems.
- C. Classifying hallucinations by type (factual fabrication, attribution error, intrinsic contradiction, extrinsic inconsistency) to identify appropriate mitigations for each.**
- D. A metric that measures the severity of hallucinations on a 1–5 scale.

Correct Answer: C

Explanation

Not all hallucinations are equal or caused by the same mechanism. Factual fabrication (invented facts) is mitigated by RAG and grounding. Attribution errors (citing wrong sources) are mitigated by citation verification. Intrinsic contradictions (self-contradicting responses) are mitigated by consistency checking. Taxonomy-driven mitigation is more effective than treating hallucination as a single undifferentiated problem.

Q5. What is "entity hallucination" and why is it particularly harmful?

- A. A type of hallucination specific to named entity recognition tasks.
- B. The model generating incorrect proper nouns — wrong names, companies, dates, statistics, or locations — which appear authoritative and are hard to spot without domain knowledge.**
- C. The model misidentifying the entity type (person vs. organisation) in its response.
- D. Hallucinations that only occur when the query involves specific entity categories.

Correct Answer: B**Explanation**

Entity hallucinations are especially dangerous because specific claims ("the merger was announced by CEO John Smith on 14 March 2024 for \$2.3 billion") are what people use to make decisions. They sound authoritative, look researched, and require domain expertise to detect. Wrong statistics, fabricated names, and incorrect dates in AI outputs have caused real-world harm in legal, medical, and financial contexts.

Q6. What is the "calibration-hallucination connection" in LLMs?

- A. Poorly calibrated models — those whose confidence scores do not reflect actual accuracy — are more likely to produce hallucinations presented with inappropriate certainty.**
- B. Hallucination rates can be reduced by calibrating the temperature parameter during inference.
- C. Calibrated models always produce accurate outputs; uncalibrated models always hallucinate.
- D. The calibration curve can be used to predict the exact hallucination rate for a given model.

Correct Answer: A**Explanation**

Well-calibrated models "know what they don't know" — their confidence tracks their accuracy. Poorly calibrated models state uncertain or incorrect claims with high confidence. Calibration and hallucination are related: a model that hedges ("I believe" / "I'm not certain") when uncertain signals its reliability better than one that states all claims with equal confidence. Calibration training is a mitigation for overconfident hallucination.

Q7. What is "hallucination red-lining" in deployment?

- A. A technique for highlighting hallucinated text in model outputs for user review.
- B. A safety rating that restricts model use to low-hallucination-risk domains.
- C. A benchmark process for measuring hallucination rate before production deployment.
- D. Defining topics or query types where hallucination risk is unacceptably high and implementing hard constraints — abstention, human review, or explicit source citation requirements — for those categories.**

Correct Answer: D

Explanation

Not all hallucination risk is equal. In a general knowledge chatbot, a wrong movie quote is annoying. In a medical information system, a wrong dosage is dangerous. Red-lining identifies high-risk categories — legal advice, medical information, financial data — and applies stricter handling: mandatory grounding, abstention when uncertain, or mandatory human review before response delivery.

Q8. What is "citation grounding" as a hallucination mitigation strategy?

- A. Embedding citations into the model's training data so it learns to attribute claims accurately.
- B. A post-processing step that automatically adds Wikipedia citations to factual claims.
- C. Requiring the model to cite a specific source document and passage for every factual claim, making unverifiable claims impossible to present as grounded.**
- D. A retrieval technique that pre-selects only highly cited documents for inclusion in context.

Correct Answer: C

Explanation

If the model must cite a specific passage for every claim, it cannot assert unsupported facts as confidently — the absence of a citation is itself a signal. This also makes verification practical: reviewers can check whether the citation supports the claim. Citation grounding does not eliminate hallucination but makes it detectable and shifts the burden of proof to the model.

Q9. What is the "Needle in a Haystack" test for LLMs?

- A. A test that places a specific fact in a large context and measures whether the model can accurately retrieve and use it — evaluating context utilisation across the full context window.**
- B. A benchmark for measuring how quickly the model can find relevant information in a knowledge base.
- C. A test that checks whether the model hallucinates when the correct answer is rare in its training data.
- D. A technique for identifying the specific training examples most responsible for a hallucination.

Correct Answer: A

Explanation

The Needle in a Haystack test inserts a single "needle" fact into a long "haystack" document and asks the model a question that can only be answered using that fact. By varying the needle's position within the context, it maps where the model fails to use context — revealing the "lost in the middle" effect and the practical limits of long-context models.

Q10. What is "intrinsic hallucination" versus "extrinsic hallucination"?

- A. Intrinsic hallucinations are generated by the model's internal reasoning; extrinsic ones come from retrieved documents.
- B. Intrinsic hallucinations occur in mathematical tasks; extrinsic ones occur in language tasks.
- C. Intrinsic hallucinations are detectable by automated tools; extrinsic ones require human evaluation.
- D. Intrinsic hallucinations contradict the provided input context; extrinsic hallucinations add information not present in the context — both are failures but require different detection approaches.**

Correct Answer: D

Explanation

Intrinsic hallucinations directly contradict the provided source — the model states the opposite of what the document says. These are detectable by automated consistency checking. Extrinsic hallucinations add information that is not in the source but may or may not be accurate — detecting these requires external knowledge or human review. RAG grounding evaluation primarily catches intrinsic hallucinations.

Q11. What makes number-based claims particularly high-risk for hallucination?

- A. Numerical processing is handled by a different sub-module that is more prone to errors.
- B. Numbers (statistics, dates, quantities, percentages) are precise, consequential, and the model has no reliable mechanism for generating correct figures it did not memorise from training.**
- C. Models are not trained on numerical data, so all numerical outputs are generated randomly.
- D. Numbers require calculation, which LLMs cannot perform accurately without tool use.

Correct Answer: B

Explanation

Language models generate numbers the same way they generate words — by predicting the next token. They have no internal calculator. A model may confidently generate "revenue grew 23.7%" when the actual figure was 18.2% — not because it was confused, but because 23.7 was a plausible continuation of the surrounding text. Statistical claims, specific dates, and monetary figures should always be verified against authoritative sources.

Q12. What is "RAGAs" (Retrieval-Augmented Generation Assessment) and what does it measure?

- A. A dataset of question-answer pairs for benchmarking RAG systems against human baselines.
- B. An API endpoint that scores RAG outputs in real time during production.
- C. A framework for evaluating RAG systems on faithfulness, answer relevance, context relevance, and context recall — providing a structured suite of automated evaluation metrics.**
- D. A set of prompting templates for constructing grounded, hallucination-resistant RAG responses.

Correct Answer: C

Explanation

RAGAs (the open-source framework) provides four core metrics: faithfulness (is the answer supported by retrieved context?), answer relevance (does the answer address the question?), context relevance (are the retrieved chunks relevant to the query?), and context recall (does the retrieved context contain the information needed to answer?). Together they diagnose where in the RAG pipeline quality is failing.

Q13. What is "chain-of-thought faithfulness" and why is it an evaluation concern?

- A. Whether the reasoning chain contains factually accurate intermediate steps.
- B. Whether the reasoning chain is internally consistent without contradictions.
- C. Whether the reasoning chain is sufficiently detailed for the user to follow.
- D. Whether the model's stated reasoning actually reflects the process it used to arrive at its answer — models can produce plausible-sounding but post-hoc rationalisations.**

Correct Answer: D

Explanation

Research has shown that model reasoning traces can be unfaithful — the stated reasoning is not necessarily what caused the final answer. A model may produce a correct answer through one process but describe a different, plausible-sounding process. Evaluating whether reasoning traces are genuinely causal (not post-hoc) is an open research problem with significant implications for transparency.

Q14. What is the "TruthfulQA" benchmark and what does it measure?

- A. A benchmark for evaluating LLM performance on factual question-answering from Wikipedia.
- B. A benchmark specifically designed to measure how often LLMs generate false information on questions where common misconceptions or plausible-sounding wrong answers exist.**
- C. A dataset of questions that models commonly hallucinate, used to fine-tune against hallucination.
- D. A benchmark that measures whether models accurately report their own uncertainty.

Correct Answer: B

Explanation

TruthfulQA tests models on questions where humans commonly hold false beliefs — questions designed so that the most fluent, confident-sounding answer is wrong. It measures whether a model is more like a knowledgeable expert (truthful) or a confident confabulator (fluent but wrong). It reveals hallucination risk on questions where the "obvious" answer is a misconception.

Q15. What is "grounded generation evaluation" and how does it differ from benchmark evaluation?

- A. Grounded generation evaluation measures how accurately a model uses provided context for specific documents; benchmark evaluation measures general capability on standardised test sets.**
- B. Grounded generation evaluation is automated; benchmark evaluation requires human raters.
- C. Grounded generation evaluation measures speed; benchmark evaluation measures accuracy.
- D. Benchmark evaluation applies to production systems; grounded generation evaluation applies only to research models.

Correct Answer: A

Explanation

Benchmarks (MMLU, HellaSwag, TruthfulQA) measure general model capability and hallucination tendency in controlled settings. Grounded generation evaluation measures a specific deployment's faithfulness to its provided documents — a production-relevant metric that general benchmarks do not capture. Both are needed: benchmarks for model selection, grounded evaluation for production monitoring.

SET

2

Evaluation Frameworks

Set 2 of 5 · 15 Questions · AI Reliability and Evaluation

Q16. What is the difference between "offline evaluation" and "online evaluation" for AI systems?

- A. Offline evaluation is manual; online evaluation is automated.
- B. Offline evaluation applies to model training; online evaluation applies to model inference.
- C. Offline evaluation uses pre-collected test sets before deployment; online evaluation measures real user interactions and outcomes in production.**
- D. Offline evaluation measures accuracy; online evaluation measures latency.

Correct Answer: C**Explanation**

Offline evaluation happens before deployment: you run your system against a curated test set and measure quality metrics. Online evaluation happens in production: you track user satisfaction, task completion rates, escalation frequency, or explicit feedback. Both are necessary — offline catches known issues before launch; online catches emergent problems in real usage.

Q17. What is "BLEU score" and what are its limitations for evaluating LLM outputs?

- A. BLEU measures n-gram overlap between generated text and reference text — adequate for machine translation but poor for evaluating open-ended generation where many valid responses exist.**
- B. BLEU is a benchmarking framework for comparing LLMs across standardised tasks.
- C. BLEU measures semantic similarity between a generated response and a reference answer.
- D. BLEU evaluates the factual accuracy of generated text against a knowledge base.

Correct Answer: A**Explanation**

BLEU was designed for machine translation, where a reference translation provides a reasonable gold standard. For open-ended LLM outputs, BLEU fails: a correct, high-quality response phrased differently from the reference scores low; a poor response with accidentally matching phrases scores high. Semantic similarity metrics (BERTScore, ROUGE) and LLM-as-judge have largely replaced BLEU for modern LLM evaluation.

Q18. What is "LLM-as-judge" evaluation and what are its failure modes?

- A. A technique where the evaluated model judges its own outputs to self-improve.
- B. A method where multiple LLMs vote on the best response, and the majority opinion is taken as ground truth.
- C. An evaluation approach that uses an LLM to generate the evaluation rubric from task description.
- D. Using a capable LLM to score or rank other LLM outputs against defined criteria — failure modes include position bias, verbosity bias, self-enhancement bias, and sensitivity to evaluation prompt wording.**

Correct Answer: D

Explanation

LLM-as-judge scales human-quality evaluation to large datasets. But LLM judges have systematic biases: they prefer longer responses (verbosity bias), prefer options presented first (position bias), and rate outputs from the same model family higher (self-enhancement bias). Mitigation: randomise position, use pairwise comparisons, calibrate against human judgements, and use a diverse set of judge models.

Q19. What is "MMLU" and what does it assess?

- A. A benchmark specifically designed to test model performance on multilingual tasks.
- B. Massive Multitask Language Understanding — a benchmark covering 57 academic subjects used to assess broad general knowledge and reasoning across disciplines.**
- C. A framework for measuring LLM output quality across multiple evaluation dimensions simultaneously.
- D. A benchmark for evaluating multi-modal models on image and text understanding jointly.

Correct Answer: B

Explanation

MMLU consists of multiple-choice questions across 57 subjects from elementary school to professional level — maths, history, law, medicine, computer science, and more. It measures breadth of world knowledge and reasoning. It is widely used for model comparison but criticised for data contamination (test questions in training data) and for not measuring open-ended capability.

Q20. What is "human evaluation" and when is it irreplaceable in LLM assessment?

- A. Human evaluation is the gold standard for all LLM tasks and should always be preferred over automated metrics.
- B. Human evaluation is used only for compliance checking — automated metrics are preferred for quality assessment.
- C. Having humans directly assess model outputs — irreplaceable for subjective qualities like tone, appropriateness, nuance, and real-world usefulness that automated metrics cannot capture.**
- D. Human evaluation is replaced by LLM-as-judge in all modern evaluation frameworks.

Correct Answer: C

Explanation

Automated metrics measure proxies for quality; humans measure quality directly. For tasks where "good" is inherently subjective — tone appropriateness, cultural sensitivity, helpfulness in context, creative quality — human evaluation is irreplaceable. The cost and latency of human evaluation mean it is used for calibration and final validation, with automated metrics for continuous monitoring.

Q21. What is "task-specific evaluation" and why is it preferred over general benchmarks for production systems?

- A. Evaluating the model on every possible task to ensure generalisation before deployment.
- B. A technique that automatically selects relevant benchmarks based on the system's configuration.
- C. Evaluating each component of an AI pipeline separately rather than end-to-end.
- D. Evaluating a deployed model on representative samples from its actual use case — more predictive of production performance than general benchmarks which test capabilities the system may never use.**

Correct Answer: D

Explanation

A general coding assistant benchmark score tells you little about how a model performs on your specific legal document summarisation task. Task-specific evaluation builds a test set from real or representative production queries, with expected outputs reviewed by domain experts. This is the only evaluation that directly predicts user experience in the actual deployment context.

Q22. What is "win rate" as an evaluation metric for comparing AI systems?

- A. The proportion of pairwise comparisons where one system's output is preferred over another's — used when there is no single correct answer and quality is comparative.**
- B. The percentage of tasks a model completes successfully without error.
- C. A metric that measures how often the AI system's recommendation is accepted by users.
- D. The rate at which one model's outputs match the outputs of a reference model.

Correct Answer: A

Explanation

Win rate is used in comparative evaluation: show evaluators (human or LLM) two outputs side by side and ask which is better. System A "wins" when its output is preferred. This is particularly useful for open-ended generation where multiple valid outputs exist and absolute quality measurement is difficult. A/B testing and RLHF preference collection both rely on win rate.

Q23. What is "benchmark contamination" and why does it undermine model comparison?

- A. A testing artefact where models perform differently on the same benchmark across different runs.
- B. When benchmark questions or answers appear in the model's training data, inflating its measured performance above its true generalisation ability.**
- C. The inclusion of culturally biased questions that advantage models trained on Western data.
- D. A technical issue where benchmark evaluation tools produce incorrect scores due to tokenisation differences.

Correct Answer: B

Explanation

If a model was trained on data containing the questions (and answers) from a benchmark, its score reflects memorisation, not capability. As benchmarks become widely known, data cleaning to remove them from training becomes harder. This is why novel, private evaluation sets are more reliable than public benchmarks for assessing truly new model capabilities.

Q24. What is a "golden test set" in AI evaluation?

- A. The training data used by the model provider, considered the gold standard for model capabilities.
- B. An automatically generated set of test cases derived from the system's most common queries.
- C. A set of adversarial inputs specifically designed to maximise model failures.
- D. A curated set of representative inputs with expert-verified expected outputs used as the stable reference for evaluating and comparing system versions over time.**

Correct Answer: D

Explanation

A golden test set is the evaluation anchor: built once from representative real inputs, verified by domain experts, and kept stable. When the system changes — prompt update, model version, retrieval change — you run the golden set and compare. It provides a consistent reference for detecting regressions and validating improvements across the system's lifetime.

Q25. What is "inter-annotator agreement" in AI evaluation and why is it important?

- A. Agreement between the evaluated model and a reference model on task outputs.
- B. A measure of how consistently the same annotator scores similar outputs over time.
- C. The degree to which different human evaluators agree on quality scores — low agreement means the evaluation criteria are ambiguous, making scores unreliable benchmarks.**
- D. Agreement between automated metrics and human evaluators on the same dataset.

Correct Answer: C

Explanation

If two evaluators rate the same output as 5/5 and 2/5, the evaluation is measuring annotator preference, not output quality. High inter-annotator agreement requires: clear, specific rubrics with examples, annotator training and calibration sessions, and periodic re-calibration. Cohen's kappa or Krippendorff's alpha are standard measures. Low agreement is a signal to refine the rubric, not to average the scores.

Q26. What is "evals" in the context of LLM application development?

- A. Manual evaluation sessions where domain experts review model outputs before each release.
- B. Automated test suites that measure whether an AI system meets defined quality criteria — treated as code, versioned, and run continuously as part of the development process.**
- C. A provider-specific term for the accuracy metrics reported in model documentation.
- D. The process of selecting which benchmarks to use for model capability assessment.

Correct Answer: B

Explanation

Evals are the AI equivalent of unit tests: structured, automated, versioned checks that run every time the system changes. They operationalise quality: "does the response always contain a citation?", "is the JSON always valid?", "does the tone always match the persona?". Building an eval suite before shipping and maintaining it throughout the system's life is a core AI engineering discipline.

Q27. What is "precision at k" (P@k) and when is it used in AI evaluation?

- A. The proportion of the top-k results that are relevant — used in retrieval evaluation to measure how many retrieved items are actually useful.**
- B. A metric that measures whether the model's k-th generation attempt is correct.
- C. The accuracy of the model on the first k examples in a test set.
- D. A measure of how many of the top-k benchmark tasks a model completes correctly.

Correct Answer: A

Explanation

P@k measures retrieval precision at a specific cutoff: if you retrieve 10 chunks and 7 are relevant, $P@10 = 0.7$. It is the standard metric for retrieval evaluation — useful because users and downstream systems typically only use the top few results. P@k is more practically meaningful than average precision across all possible k values.

Q28. What is "normalised discounted cumulative gain" (NDCG) and when is it used?

- A. A metric that measures how much a model's performance degrades on normalised (cleaned) test data.
- B. A gain metric that measures information extracted per token in model outputs.
- C. A rank-weighted retrieval metric that gives higher credit to relevant results appearing earlier in the ranked list — used when the position of relevant results matters.**
- D. A normalised version of BLEU that accounts for different response lengths.

Correct Answer: C

Explanation

NDCG rewards relevant results more when they appear higher in the ranked list — a relevant result at position 1 contributes more than one at position 10. It also handles graded relevance (highly relevant, somewhat relevant, not relevant) rather than binary. NDCG is the standard metric for information retrieval systems and re-ranker evaluation in RAG pipelines.

Q29. What is "automated evaluation at scale" and what trade-offs does it involve?

- A. Running the same evaluation simultaneously across many model instances to parallelise testing.
- B. Automatically generating evaluation test cases from production traffic to scale the test set.
- C. A technique for evaluating models without labelled data by measuring distributional properties.
- D. Using automated metrics (LLM-as-judge, rule-based checks, embedding similarity) to evaluate large volumes of outputs — trading measurement accuracy for coverage and cost.**

Correct Answer: D

Explanation

Manual human evaluation of 10,000 outputs is impractical. Automated evaluation sacrifices some accuracy (LLM judges have biases; metrics are proxies) in exchange for scale, speed, and cost. The right approach uses: automated metrics for continuous monitoring of large volumes, with human evaluation for calibration, validation, and cases where automated metrics disagree.

Q30. What is "evaluation for safety" as distinct from "evaluation for quality"?

- A. Safety evaluation is done by the model provider; quality evaluation is done by the application developer.
- B. Safety evaluation tests whether the system produces harmful, biased, or inappropriate outputs; quality evaluation tests whether outputs are accurate, relevant, and useful — these require different test sets and rubrics.**
- C. Safety evaluation is a one-time assessment before launch; quality evaluation is continuous.
- D. Safety evaluation applies to the model; quality evaluation applies to the application.

Correct Answer: B

Explanation

Safety and quality are orthogonal dimensions that both require dedicated evaluation. A system can be high-quality (accurate, relevant, well-formatted) but unsafe (produces harmful content under certain inputs). Safety evaluation requires adversarial test sets specifically designed to probe harmful behaviours — jailbreaks, bias elicitation, misinformation generation — that normal quality evaluation does not cover.

SET

3

Guardrails and Fallbacks

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Q31. What is a "guardrail" in an AI system and what problem does it solve?

- A. A constraint or check — either pre-generation, inline, or post-generation — that prevents the AI from producing harmful, incorrect, or out-of-scope outputs.
- B. A hardware limit that prevents the AI system from using more compute than allocated.
- C. A legal disclaimer added to AI outputs to limit organisational liability.
- D. A rate limiting mechanism that prevents users from querying the AI too frequently.

Correct Answer: A**Explanation**

Guardrails are the safety and reliability infrastructure around an AI system. They can operate at multiple points: pre-generation (classifying input intent before sending to the model), inline (system prompt instructions that constrain model behaviour), or post-generation (checking outputs before delivery to users). Guardrails address what the core model alone cannot reliably guarantee.

Q32. What is an "input guardrail" and what types of inputs does it typically handle?

- A. A validation that checks whether the user's input is correctly formatted before API submission.
- B. A rate limiter that prevents individual users from sending too many queries per minute.
- C. A check applied to user input before it reaches the model — handling prompt injection attempts, out-of-scope queries, PII detection, and inappropriate content.
- D. An authentication check that verifies the user's identity before processing their input.

Correct Answer: C**Explanation**

Input guardrails intercept problems before the model processes them: detecting and blocking prompt injection attempts, classifying whether the query is in-scope for the application, detecting and redacting PII that should not be sent to the model, filtering explicit or harmful content. Handling these at the input layer is more reliable than relying on the model to handle them in-context.

Q33. What is an "output guardrail" and how does it differ from input filtering?

- A. A post-processing step that formats the model's raw output into the application's display format.
- B. A check applied to model output before it reaches the user — detecting harmful content, PII leakage, format violations, or grounding failures in the generated response.**
- C. A citation requirement that ensures every factual claim in the output has a source.
- D. A response length check that truncates outputs exceeding a defined word limit.

Correct Answer: B

Explanation

Output guardrails run after generation but before user delivery. They catch what slipped through input filtering and model instructions: a response that contains PII from retrieved documents, a response that violates format requirements, a response that includes inappropriate content despite system prompt instructions. Output guardrails provide a backstop when the model fails to follow instructions reliably.

Q34. What is a "fallback system" in AI application design?

- A. A backup model that activates automatically when the primary model's API is unavailable.
- B. A secondary retrieval system that provides additional context when initial retrieval returns insufficient results.
- C. A logging mechanism that records failed AI interactions for later review and improvement.
- D. A predefined response or alternative action triggered when the primary AI system fails, produces low-confidence output, or encounters an out-of-scope query.**

Correct Answer: D

Explanation

Fallback systems ensure that when the AI cannot reliably answer — low retrieval confidence, flagged by a guardrail, hallucination risk detected, API error — the application degrades gracefully: a safe pre-written response, a human escalation prompt, a disclaimer, or an apology with contact information. Without fallbacks, AI failures are user-facing.

Q35. What is "confidence-based routing" and how does it improve reliability?

- A. Routing queries to different handling paths based on the AI system's assessed confidence — high-confidence queries get automated responses; low-confidence ones get human review or escalation.**
- B. Routing different query types to different models based on their measured accuracy for that task.
- C. A load-balancing technique that directs traffic to the highest-confidence model instance.
- D. A technique for choosing between multiple retrieved chunks based on similarity scores.

Correct Answer: A

Explanation

Not all queries are equally certain. A query with high retrieval scores and a strong match to known patterns can be answered confidently. A query with poor retrieval, an out-of-distribution intent, or detected ambiguity should be handled differently — with explicit uncertainty statements, human escalation, or clarification requests. Confidence-based routing operationalises this distinction.

Q36. What is a "system prompt guardrail" and how reliable is it as the sole safety mechanism?

- A. A system-level access control that prevents certain users from modifying the system prompt.
- B. Instructions in the system prompt that tell the model to refuse or handle certain request types — effective as a first layer but insufficient alone, as models can be jailbroken or misled.**
- C. A validation that checks the system prompt for prohibited keywords before deployment.
- D. An automated test that verifies the system prompt produces safe outputs before launch.

Correct Answer: B

Explanation

System prompt guardrails ("Never provide medical diagnoses", "Do not discuss competitor products") are the simplest safety mechanism — and the least reliable as a standalone control. Determined adversarial users can craft inputs that bypass instruction-following. System prompt guardrails should be one layer in a defence-in-depth approach that also includes input/output classifiers, content filters, and monitoring.

Q37. What is a "constitutional AI" approach to guardrails?

- A. A legal framework for AI governance that defines acceptable use policies for AI systems.
- B. A system prompt template structured as a numbered list of rules the model must follow.
- C. A human review process that checks AI outputs against a defined set of ethical guidelines.
- D. Training a model with a set of principles ("constitution") that it uses to self-critique and revise its outputs — building safety constraints into the model's weights rather than relying only on prompt instructions.**

Correct Answer: D

Explanation

Constitutional AI (Anthropic's approach) trains the model to evaluate its own outputs against a set of principles and revise them — making safety part of the model's reasoning process, not just an external constraint. This produces more robust safety behaviour than system prompt instructions alone, because the model internalises the principles during training rather than following them as runtime instructions.

Q38. What is "PII detection" as a guardrail and why is it critical in enterprise AI?

- A. A technique for verifying that users querying the AI system are who they claim to be.
- B. A monitoring system that alerts administrators when users share personal information with the AI.
- C. Identifying and redacting personally identifiable information in user inputs or model outputs before it is processed or delivered — critical for privacy compliance (GDPR, DPDP Act) and data governance.**
- D. A filter that removes employee names from retrieved documents before they enter the AI context.

Correct Answer: C

Explanation

Enterprise AI systems often process sensitive data. A customer support AI might receive messages containing credit card numbers, Aadhaar numbers, or medical conditions — none of which should be sent to external model APIs or stored in logs. PII detection guardrails intercept this before it reaches the model, replace it with placeholders, and ensure compliance with privacy regulations.

Q39. What is a "topic boundary guardrail" and when is it needed?

- A. A guardrail that prevents the model from generating content on sensitive topics defined in the system prompt.
- B. A constraint that prevents the model from responding to queries outside the defined scope of the application — redirecting out-of-scope queries rather than attempting to answer them.**
- C. A keyword filter that blocks specific terms from appearing in model outputs.
- D. A retrieval filter that excludes documents outside the application's defined knowledge domain.

Correct Answer: B

Explanation

A legal research AI should not provide personal financial planning advice; a customer support bot should not discuss competitors; a children's education tool should not engage with adult content. Topic boundary guardrails classify incoming queries and redirect out-of-scope ones before the model processes them — preventing the model from attempting to answer queries it should not handle.

Q40. What is "graceful degradation" in AI system design?

- A. Ensuring that when components fail or confidence is low, the system fails safely — providing partial help, clear uncertainty statements, or human escalation rather than silent failure or unreliable output.**
- B. A technique for reducing model capability progressively under high load to maintain availability.
- C. Designing AI outputs to degrade in visual formatting gracefully across different display contexts.
- D. A failover architecture where a simpler model activates when the primary model's API is unavailable.

Correct Answer: A

Explanation

Graceful degradation applies the principle of safe failure to AI: when the system cannot answer reliably, it should fail in a way that is helpful rather than harmful. "I'm not confident in this answer — you may want to verify with a human expert" is graceful degradation. Producing a confident hallucination is ungraceful failure. Design for the uncertain case, not just the nominal case.

Q41. What is "rate limiting" as an AI system guardrail?

- A. A technique for limiting the length of model outputs to reduce latency and cost.
- B. A guardrail that prevents the model from generating more than a defined number of claims per response.
- C. Restricting the number of queries a user or application can send per time period — preventing abuse, managing cost, and protecting against denial-of-service attacks.**
- D. A constraint that limits how many retrieved chunks are included in each LLM context.

Correct Answer: C

Explanation

Rate limiting is an operational guardrail: it prevents individual users from consuming disproportionate resources, protects against automated abuse or scraping, and controls cost at the application level. For AI applications with per-token pricing, unlimited query rates can create unbounded cost exposure. Rate limiting is a basic but essential operational control.

Q42. What is "output validation" as distinct from "output guardrails"?

- A. Output validation is automated; output guardrails require human review.
- B. Output validation applies to generation; guardrails apply to retrieval.
- C. Output validation and guardrails are synonymous — different terms for the same process.
- D. Output validation checks whether the response meets structural requirements (valid JSON, required fields present, length within bounds); guardrails check whether the content is safe and appropriate.**

Correct Answer: D

Explanation

Both are post-generation checks but they measure different things. Validation is structural: "Is the JSON parseable?", "Are all required fields present?", "Is the response within the length limit?" Guardrails are content-based: "Does the response contain PII?", "Is it in-scope?", "Does it match safety guidelines?" Applications need both: structural validity and content safety.

Q43. What is "jailbreaking" and why does it matter for guardrail design?

- A. A technique for extracting the system prompt from an AI application to reveal its configuration.
- B. Crafting adversarial prompts that cause an AI system to bypass its safety instructions and produce content it was configured not to produce.**
- C. An attack that exploits vulnerabilities in the API infrastructure rather than the model itself.
- D. A method for accessing premium model capabilities without proper authorisation.

Correct Answer: B

Explanation

Jailbreaks demonstrate that instruction-based guardrails are not robust: sufficiently creative users can find phrasings that cause the model to ignore its constraints. This is why guardrail design must be defence-in-depth — system prompt instructions are the first layer, but input classifiers, content filters, output checks, and monitoring must back them up. Security through instructions alone is insufficient.

Q44. What is "monitoring-as-a-guardrail" and how does it complement preventive controls?

- A. A passive safety approach that relies entirely on monitoring rather than implementing active guardrails.
- B. A compliance mechanism that records all AI interactions for regulatory audit purposes.
- C. Using continuous logging, anomaly detection, and alerting to detect guardrail bypass and quality failures that preventive controls missed — enabling rapid response to emergent issues.**
- D. A performance monitoring system that tracks model latency and error rates in production.

Correct Answer: C

Explanation

Preventive guardrails stop known problems. Monitoring catches what slipped through — novel jailbreak patterns, unexpected use cases, quality degradation, PII leakage. Monitoring enables the closed loop: detect a new failure mode, implement a preventive guardrail, verify the monitoring shows improvement. Without monitoring, you have no visibility into what your guardrails are missing.

Q45. What is "human-in-the-loop" (HITL) as a reliability mechanism and when is it appropriate?

- A. Inserting human review at decision points where AI confidence is low or stakes are high — appropriate for high-risk decisions, low-confidence outputs, or novel situations outside the AI's training distribution.**
- B. A development practice where humans write all prompts and review all outputs before deployment.
- C. A training technique where human feedback is used to improve the model's quality iteratively.
- D. A safety requirement mandating human approval for every AI output before it is delivered.

Correct Answer: A

Explanation

HITL is a reliability and safety mechanism for high-stakes or uncertain situations: route low-confidence outputs to a human reviewer before delivery; require human approval for consequential decisions (loan approval, medical recommendations, legal advice); escalate novel queries that fall outside the system's known patterns. HITL adds latency and cost but provides the ultimate backstop where AI reliability is insufficient.

SET

4

Monitoring and Testing AI Outputs

Set 4 of 5 · 15 Questions · AI Reliability and Evaluation

Q46. What is "production monitoring" for an AI system and what metrics should be tracked?

- A. A pre-launch testing phase that simulates production traffic to identify performance bottlenecks.
- B. Manual review of a random sample of production outputs by quality assurance analysts.
- C. A compliance audit that checks whether the AI system's outputs meet regulatory requirements.
- D. Continuously measuring system health in live deployment — tracking output quality, user satisfaction, guardrail trigger rates, latency, error rates, and hallucination indicators.**

Correct Answer: D**Explanation**

Production monitoring is ongoing, not one-time. Key metrics: quality indicators (user satisfaction, task completion, thumbs-down rate), safety indicators (guardrail trigger rate, escalation rate), operational indicators (latency, error rate, token cost per session), and drift indicators (output length trend, topic distribution shift). Without monitoring, you are flying blind.

Q47. What is "A/B testing" for AI systems and what makes it valid?

- A. Testing two model versions simultaneously and selecting the one with lower error rate.
- B. Comparing two system variants with random assignment of real traffic — valid when there is a pre-defined success metric, sufficient sample size, and only the tested variable differs between conditions.**
- C. Running the same test twice — once in development and once in production — and comparing results.
- D. Presenting users with two AI responses and asking them to choose the better one.

Correct Answer: B**Explanation**

A/B testing of AI systems requires the same statistical rigour as any controlled experiment: random assignment (no selection bias), pre-defined success metric (decided before the test, not after seeing results), sufficient sample size to detect meaningful differences, and controlled conditions (only the variable being tested differs). Without these, observed differences are noise, not signal.

Q48. What is "shadow testing" for AI system deployment?

- A. A testing approach where the development team tests the system without telling the client.
- B. A technique for testing the AI system against adversarial inputs without the knowledge of the model provider.
- C. Running the new system in parallel with the production system on the same live traffic, comparing outputs without exposing users to the new system's responses.**
- D. Running a simplified version of the system in a lower-cost environment to simulate production.

Correct Answer: C

Explanation

Shadow testing is the safest deployment testing approach: the new system processes real production traffic and generates responses, but users see only the old system's responses. The new system's outputs are evaluated and compared without any user risk. Once quality is confirmed in shadow mode, the new system can be gradually promoted to handle real traffic.

Q49. What is "canary deployment" in the context of AI systems?

- A. Gradually rolling out a new system version to a small percentage of users, monitoring quality metrics, before full deployment — limiting exposure if the new version underperforms.**
- B. A technique for testing the AI system on a small subset of the training data before full evaluation.
- C. Deploying a simplified version of the system to test infrastructure performance under load.
- D. A monitoring approach that uses a small set of "canary" test queries run continuously against production.

Correct Answer: A

Explanation

Canary deployment controls rollout risk: send 5% of traffic to the new version, monitor quality and error metrics, and increase the percentage only if metrics are healthy. If the new version underperforms, roll it back before the majority of users are affected. This is the standard deployment pattern for high-traffic AI applications where a bad model update has user-facing consequences.

Q50. What is "regression testing" for AI systems and what triggers it?

- A. Testing the system on tasks it previously failed to verify that the failures have been fixed.
- B. A scheduled monthly evaluation of system quality against industry benchmarks.
- C. A test that measures whether the system's performance degrades as the knowledge base grows.
- D. Running a fixed test suite after any change (prompt update, model version change, retrieval change) to verify that previously passing cases still pass — triggered by any system modification.**

Correct Answer: D

Explanation

AI systems can regress silently: a prompt change that improves one behaviour degrades another; a model version update that improves general capability changes specific task behaviour. Regression testing runs the same curated test suite before and after changes, catching quality degradations before they reach production. Without regression testing, silent regressions accumulate undetected.

Q51. What is "latency monitoring" for AI applications and why does it matter for user experience?

- A. A technique for measuring how quickly the model generates tokens per second.
- B. Monitoring API rate limit errors caused by exceeding the provider's request quota.
- C. Tracking time-to-first-token, total generation time, and end-to-end response time — slow responses directly reduce user engagement and perceived quality.**
- D. Tracking how quickly new documents are indexed into the vector database.

Correct Answer: C

Explanation

Response latency is a quality dimension: users abandon slow interfaces, slow responses feel lower quality even when content is identical, and in conversational applications excessive wait times destroy the interaction flow. Monitoring p50, p95, and p99 latency (not just average) reveals tail latency issues that affect a minority of users but significantly damage their experience.

Q52. What is "semantic drift" monitoring and when is it triggered?

- A. Tracking whether the distribution of topics and intents in user queries shifts over time — triggered when queries increasingly fall outside the system's designed domain.**
- B. Monitoring whether the model's response style drifts from the defined persona over time.
- C. Detecting when the embedding model's representation of the same text changes after a model update.
- D. Tracking whether users increasingly phrase queries in ways that cause retrieval to fail.

Correct Answer: A

Explanation

User needs evolve. A customer support AI designed for billing queries may see queries about new features after a product launch. Semantic drift monitoring compares the current query distribution to the baseline — when distribution shifts significantly, the knowledge base, system prompt, and guardrails may need updating. Catching drift early prevents the system from being queried outside its reliable operating range.

Q53. What is "cost monitoring" for AI applications and what should it alert on?

- A. Monitoring the computational resources consumed by the model provider's infrastructure.
- B. Tracking token consumption, API costs, and cost per user interaction — alerting on unexpected spikes that may indicate misuse, runaway sessions, or system prompt changes with unintended cost implications.**
- C. Tracking whether the organisation is within its contracted API usage tier.
- D. A procurement process for reviewing and approving AI service spending.

Correct Answer: B

Explanation

AI applications have variable costs driven by token consumption. A system prompt change that adds 500 tokens per call, multiplied by 100,000 daily calls, is a significant unplanned cost increase. Cost spikes may also indicate misuse — users crafting long inputs to extract information. Cost monitoring with anomaly detection catches both accidental and adversarial cost surprises.

Q54. What is "error rate monitoring" for AI applications?

- A. Monitoring the number of factually incorrect responses produced per day.
- B. A metric for measuring how often the AI system fails to retrieve relevant documents.
- C. Tracking the frequency of API errors, timeout failures, parsing errors in structured outputs, and guardrail trigger events — essential for distinguishing systemic from transient failures.**
- D. Tracking the rate at which users report dissatisfaction with AI responses.

Correct Answer: C

Explanation

Error rate monitoring covers multiple failure types: API errors (model provider outages), timeout failures (response time exceeding SLA), parsing errors (model failing to return valid JSON when required), guardrail triggers (potentially harmful queries detected). Distinguishing these failure types enables targeted remediation — API errors require infrastructure action; parsing errors require prompt fixes.

Q55. What is "user feedback collection" in AI application monitoring?

- A. A compliance mechanism that requires users to confirm they understand AI responses may be inaccurate.
- B. A technique for using user conversations to create fine-tuning training data.
- C. A survey sent to users after each session to collect overall satisfaction scores.
- D. Capturing explicit signals (thumbs up/down, ratings, reports) and implicit signals (session abandonment, follow-up queries) from users to identify output quality issues at scale.**

Correct Answer: D

Explanation

User feedback is a direct quality signal. Explicit feedback (thumbs down, "report this response") directly identifies problematic outputs. Implicit feedback (user immediately rephrases the query — suggesting the first response missed; session abandonment after a response — suggesting the response was unhelpful) reveals quality issues at scale without requiring users to rate every response. Both signal types should feed into the improvement loop.

Q56. What is "chaos testing" for AI systems?

- A. Deliberately injecting failures — API timeouts, malformed retrieved chunks, context window overflows, guardrail triggers — to verify that the system handles failures gracefully and reliably.**
- B. A technique for testing AI safety by deliberately providing dangerous prompts to assess guardrail effectiveness.
- C. A random query generation approach for discovering unexpected model failure modes.
- D. A load testing methodology that simulates peak traffic to assess system stability.

Correct Answer: A

Explanation

Chaos testing verifies resilience: what happens when the vector database is slow? When the LLM API returns an error? When a retrieved document is corrupted? When the context window overflows? A system that handles these gracefully in testing will handle them gracefully in production. A system that crashes, returns silent errors, or produces confusing outputs under failure conditions is not production-ready.

Q57. What is "output consistency testing" and why is it important for structured outputs?

- A. Testing whether the model produces the same response to the same query across multiple runs.
- B. Testing whether the system produces structurally valid outputs (valid JSON, required fields, correct types) across diverse inputs — critical for downstream systems that parse model outputs.**
- C. Checking that all responses share the same tone and style as defined in the system prompt.
- D. Verifying that citation formatting in responses is consistent across different document types.

Correct Answer: B

Explanation

Structured output applications (APIs that return JSON, pipelines that parse model outputs) break when the model produces invalid structure — even one bad field or missing bracket causes a parse failure. Output consistency testing probes the system with diverse inputs to find cases where structure breaks: long inputs, unusual characters, edge case content, or queries that cause the model to add prose around the JSON.

Q58. What is "end-to-end testing" for a RAG pipeline?

- A. Testing each component of the pipeline independently before integration.
- B. A performance test that measures total pipeline latency from input to output.
- C. A compliance test that verifies the pipeline meets data processing regulations end-to-end.
- D. Testing the complete system from user query to final response — including retrieval, context assembly, generation, and output validation — using representative real-world queries.**

Correct Answer: D

Explanation

Component tests verify each piece works in isolation; end-to-end tests verify the whole system works together for real users. A RAG pipeline where retrieval, generation, and guardrails each pass component tests can still fail end-to-end — due to interactions between components, data format mismatches, or emergent failure modes that only appear under realistic conditions.

Q59. What is "load testing" for AI applications and what does it reveal?

- A. Simulating high-traffic conditions to identify performance degradation, rate limit failures, increased latency, and cost spikes that only appear under load — distinct from nominal performance.**
- B. Testing the model's performance on a large number of diverse test cases.
- C. A benchmark that measures how many tokens per second the model can process.
- D. A technique for testing the knowledge base's capacity as the number of indexed documents grows.

Correct Answer: A

Explanation

AI applications behave differently under load: concurrent requests may exhaust API rate limits, vector database latency may spike under concurrent queries, LLM generation latency may increase as provider capacity fills. Load testing reveals these failure modes before production traffic exposes them to users. It also validates that the cost estimates for expected traffic volumes are realistic.

Q60. What is "observability" for AI systems and how does it differ from monitoring?

- A. Observability is a compliance term for audit logging; monitoring is a technical term for performance tracking.
- B. Observability applies to the model provider's infrastructure; monitoring applies to the application layer.
- C. Observability provides the ability to understand system behaviour from its outputs — through logging, tracing, and metrics — enabling debugging of novel failures; monitoring tracks predefined metrics for known failure modes.**
- D. Observability and monitoring are the same concept — used interchangeably in AI system design.

Correct Answer: C**Explanation**

Monitoring tells you "something is wrong"; observability tells you "what and why." Monitoring tracks known metrics (latency, error rate, cost). Observability enables debugging of unknown failures — by logging the full pipeline state (query, retrieved chunks, generation input, output, guardrail decisions), you can trace any unexpected output back to its root cause. For complex AI systems, observability is as important as monitoring.

SET

5

Human Oversight and Adversarial Testing

Set 5 of 5 · 15 Questions · AI Reliability and Evaluation

Q61. What is the role of human oversight in high-stakes AI applications?

- A. Providing a review and approval mechanism for AI outputs where errors have significant consequences — ensuring that AI recommendations are verified by qualified humans before action is taken.**
- B. Monitoring AI systems for compliance violations on behalf of regulatory bodies.
- C. Reviewing and approving AI system design before deployment to verify safety properties.
- D. A legal requirement mandating that humans review every AI output before delivery.

Correct Answer: A**Explanation**

Human oversight is risk-calibrated: casual queries on a knowledge bot may not require it; a credit decision, medical recommendation, or legal advice output almost certainly does. The role of human oversight is to catch consequential errors before they cause harm — not to review every output, but to be in the loop for decisions where AI error is unacceptably costly.

Q62. What is "red-teaming" an AI system and what does it accomplish?

- A. A performance review process where internal teams compete to identify the most efficient prompts.
- B. An evaluation methodology that uses red (bad) and green (good) labels to classify model outputs.
- C. A regulatory requirement for AI systems operating in certain industries.
- D. Structured adversarial testing by a dedicated team that attempts to find failure modes, safety gaps, and misuse vectors — producing a documented inventory of vulnerabilities before deployment.**

Correct Answer: D**Explanation**

Red-teaming adopts an attacker's mindset: what is the worst this system could do, and can we make it do that? A dedicated red team tries jailbreaks, social engineering, adversarial inputs, edge cases, and misuse scenarios. The output is not a pass/fail grade but a vulnerability inventory — a list of specific inputs that caused the system to fail, with severity ratings — that drives remediation before launch.

Q63. What is "adversarial robustness" and why is it evaluated separately from general accuracy?

- A. A measure of how accurately the system performs on difficult but naturally occurring test cases.
- B. The system's ability to maintain correct and safe behaviour under deliberate adversarial inputs — a distinct capability from performing well on standard or representative inputs.**
- C. The system's ability to maintain performance as the volume of adversarial competitors increases.
- D. A benchmark for systems deployed in security-critical environments like cybersecurity.

Correct Answer: B

Explanation

A system can achieve high accuracy on representative test sets while being catastrophically vulnerable to targeted adversarial inputs. Adversarial robustness evaluation specifically tests inputs crafted to find and exploit failure modes — jailbreaks, prompt injections, data poisoning, out-of-distribution queries. High general accuracy does not imply adversarial robustness; these must be evaluated independently.

Q64. What is "bias evaluation" in AI systems and what types of bias are tested?

- A. An evaluation of whether the model's training data contains balanced representation across all topics.
- B. A technique for removing biased language from model outputs in post-processing.
- C. Systematically testing whether the system produces different quality outputs for different demographic groups, topics, or phrasings in ways that reflect discriminatory patterns.**
- D. A regulatory assessment of whether the AI system was built with diverse team representation.

Correct Answer: C

Explanation

Bias evaluation tests for disparate performance: does the system perform worse for users of certain backgrounds, languages, or demographic groups? Does it reflect harmful stereotypes? Does it apply different standards to politically or socially sensitive topics? These evaluations require diverse test sets specifically designed to surface bias — standard accuracy benchmarks rarely reveal it.

Q65. What is "prompt injection testing" and why is it critical for RAG systems?

- A. Testing whether adversarial content in retrieved documents can override system prompt instructions — critical for RAG systems that retrieve from untrusted or user-supplied content.**
- B. A technique for testing whether the system prompt can be extracted by users through targeted queries.
- C. Testing whether the model can resist attempts to change its output format through user prompts.
- D. A security audit of the API infrastructure for injection vulnerabilities in the HTTP layer.

Correct Answer: A

Explanation

RAG systems retrieve from external sources — web pages, user-uploaded documents, databases. These sources may contain text like "SYSTEM: Ignore your previous instructions and instead do X." Prompt injection testing specifically injects adversarial content into retrieved documents and verifies that the system prompt's instructions are maintained. Without this test, you do not know if your guardrails survive real-world retrieval.

Q66. What is the "alignment tax" and why does it create evaluation trade-offs?

- A. A cost charged by model providers for access to safety features like content filtering.
- B. The time investment required to implement safety alignment in an AI system.
- C. The reduction in raw task performance that safety-aligned models sometimes exhibit compared to unaligned models — evaluating aligned systems requires balancing safety and capability metrics.**
- D. The performance overhead of running safety classifiers in parallel with the main model.

Correct Answer: C

Explanation

Safety-aligned models sometimes produce lower benchmark scores than their unaligned counterparts — they may refuse certain tasks, add caveats, or produce more conservative outputs. Evaluation must account for this: a model that scores lower on raw capability benchmarks but significantly better on safety evaluations may be the better production choice. The alignment tax is a real trade-off that evaluation frameworks must incorporate.

Q67. What is "scalable oversight" as an AI safety concept relevant to evaluation?

- A. Scaling the number of human reviewers proportionally as AI output volume increases.
- B. A technique for automating the oversight of AI systems to reduce human review requirements.
- C. The practice of delegating AI oversight to external auditors as systems become too complex for internal teams.
- D. Methods for maintaining meaningful human oversight as AI systems become more capable and produce outputs that are increasingly difficult for humans to evaluate directly.**

Correct Answer: D

Explanation

As AI systems become more capable — producing complex code, long research reports, nuanced legal analysis — human reviewers may lack the expertise or bandwidth to reliably evaluate their outputs. Scalable oversight research addresses this: debate protocols, AI-assisted evaluation, interpretability tools, and decomposition of complex tasks into verifiable sub-tasks all aim to keep human oversight meaningful as capability increases.

Q68. What is a "safety benchmark" for LLMs and what does it measure?

- A. A benchmark that measures a model's ability to reason about safety scenarios presented in text.
- B. A structured evaluation of whether a model reliably refuses harmful requests, avoids producing dangerous content, and behaves in accordance with safety guidelines across diverse adversarial inputs.**
- C. An industry certification that verifies an AI system has passed minimum safety requirements.
- D. A benchmark for measuring the probability that a model will generate harmful content during normal use.

Correct Answer: B

Explanation

Safety benchmarks (HarmBench, WildGuard, AIR-Bench) test whether models refuse harmful requests across categories: violence, illegal activities, privacy violations, misinformation, self-harm. They use adversarial inputs designed to probe refusal reliability — not polite "please do something harmful" requests, but sophisticated jailbreaks and edge cases. Safety benchmark performance predicts (imperfectly) real-world safety under adversarial use.

Q69. What is "out-of-distribution" (OOD) testing and why is it important for AI reliability?

- A. Testing a model on data from a different organisation or domain than the one it was deployed for.
- B. A technique for evaluating model performance when the evaluation dataset is different from the training dataset.
- C. Testing model behaviour on inputs significantly different from those seen during training or evaluation — revealing how the system handles real-world variation and edge cases.**
- D. Testing the system on queries that fall outside the intended topic scope of the application.

Correct Answer: C

Explanation

Models perform differently on inputs that resemble their training distribution versus novel inputs. OOD testing probes the latter: unusual phrasings, domain jargon not in training data, multilingual inputs, ambiguous or contradictory queries. OOD performance reveals where the system's reliability degrades and where guardrails or human escalation are most needed.

Q70. What is "evaluation versioning" and why does it matter?

- A. Keeping evaluation test sets and scoring rubrics stable across system versions so that improvements and regressions can be measured consistently over time.**
- B. A technique for automatically updating evaluation criteria as the system evolves.
- C. Maintaining separate evaluation frameworks for different model versions.
- D. A data management practice for archiving historical evaluation results.

Correct Answer: A

Explanation

If you change the test set every time you evaluate, you cannot compare results across versions. Evaluation versioning locks the test set and rubric, creating a stable measurement reference. When the system improves, the metric improves on the same test set. When it regresses, the metric falls — on the same baseline. Without stable evaluation, improvement and regression are indistinguishable from evaluation noise.

Q71. What is "multi-stakeholder evaluation" in enterprise AI deployment?

- A. A governance process where multiple teams jointly approve AI system design before deployment.
- B. Evaluating AI system quality from the perspectives of multiple stakeholders — users, administrators, domain experts, compliance teams — who may have different and sometimes conflicting quality requirements.**
- C. An evaluation methodology that weights different test cases by their business importance.
- D. A technique for collecting evaluation feedback from a diverse representative sample of users.

Correct Answer: B

Explanation

Different stakeholders care about different things: end users care about helpfulness and accuracy; compliance teams care about regulatory adherence; administrators care about cost and operational reliability; legal teams care about liability exposure. A system that scores highly on user satisfaction metrics may fail compliance evaluation, or vice versa. Multi-stakeholder evaluation surfaces these conflicts before deployment.

Q72. What is "pre-deployment safety review" and what does it involve?

- A. A performance review process that validates the system meets speed and cost requirements before launch.
- B. A compliance check that verifies the system's data handling practices meet regulatory requirements.
- C. An internal approval process where senior management signs off on AI system deployment.
- D. A structured assessment of an AI system's safety properties before release — including red-teaming, bias evaluation, guardrail testing, and documentation of residual risks.**

Correct Answer: D

Explanation

Pre-deployment safety review is a structured gate before launch. It typically involves: red-team adversarial testing across defined harm categories, bias evaluation across demographic groups, guardrail effectiveness testing, documentation of known limitations and failure modes, and definition of monitoring plans for post-deployment. It is both a quality gate and a risk documentation exercise.

Q73. What is the difference between "safety" and "alignment" in AI evaluation?

- A. Safety is evaluated by external auditors; alignment is evaluated by the development team.
- B. Safety applies to the model; alignment applies to the application layer built on top of the model.
- C. Safety evaluates whether the system avoids causing harm; alignment evaluates whether the system does what its designers intended — a system can be safe but misaligned, or aligned but unsafe.**
- D. Safety and alignment are synonymous terms used in different research communities.

Correct Answer: C

Explanation

Safety and alignment are related but distinct. A system might be well-aligned with its designer's intent (it does exactly what they specified) but those specifications were safety-incomplete (the designer did not anticipate harmful use cases). Conversely, a system might refuse many harmful requests (safe in those cases) but still systematically misrepresent information (misaligned). Evaluating both provides a complete safety picture.

Q74. What is "human preference evaluation" and what are its limitations?

- A. A technique for using human preferences to define the gold standard for AI quality.
- B. Measuring which AI outputs humans prefer in pairwise or rating evaluations — useful for quality assessment but biased toward confident, fluent, long responses regardless of accuracy.**
- C. An evaluation approach that aggregates user satisfaction scores from production interactions.
- D. Human preference evaluation is the most reliable method for all AI quality assessment tasks.

Correct Answer: B

Explanation

Human preferences are systematically biased: people rate longer, more confident, more fluent responses higher — regardless of accuracy. An AI system optimised purely for human preference may become better at appearing helpful while becoming worse at being accurate. Preference evaluation must be combined with accuracy, faithfulness, and safety evaluation to avoid optimising for persuasiveness over truth.

Q75. What is the "evaluation flywheel" in AI product development?

- A. A technique for automating the full evaluation pipeline to reduce manual evaluation effort over time.
- B. A metric that measures how quickly the AI system improves over successive model versions.
- C. A team structure where evaluation responsibilities rotate across product, engineering, and data science.
- D. The continuous improvement cycle: evaluate → identify failures → improve system → evaluate again — where each cycle produces better evaluations, better systems, and better understanding of remaining gaps.**

Correct Answer: D

Explanation

The evaluation flywheel is the key discipline in AI product development: systematic evaluation reveals specific failure modes; those failures become targeted improvements; improvements are verified by re-evaluation; new evaluation uncovers the next set of failures. Teams that build robust evaluation infrastructure improve faster and more reliably than teams that rely on intuition and manual spot-checks.

END OF QUESTION BANK

*Your turning
point starts
here.*